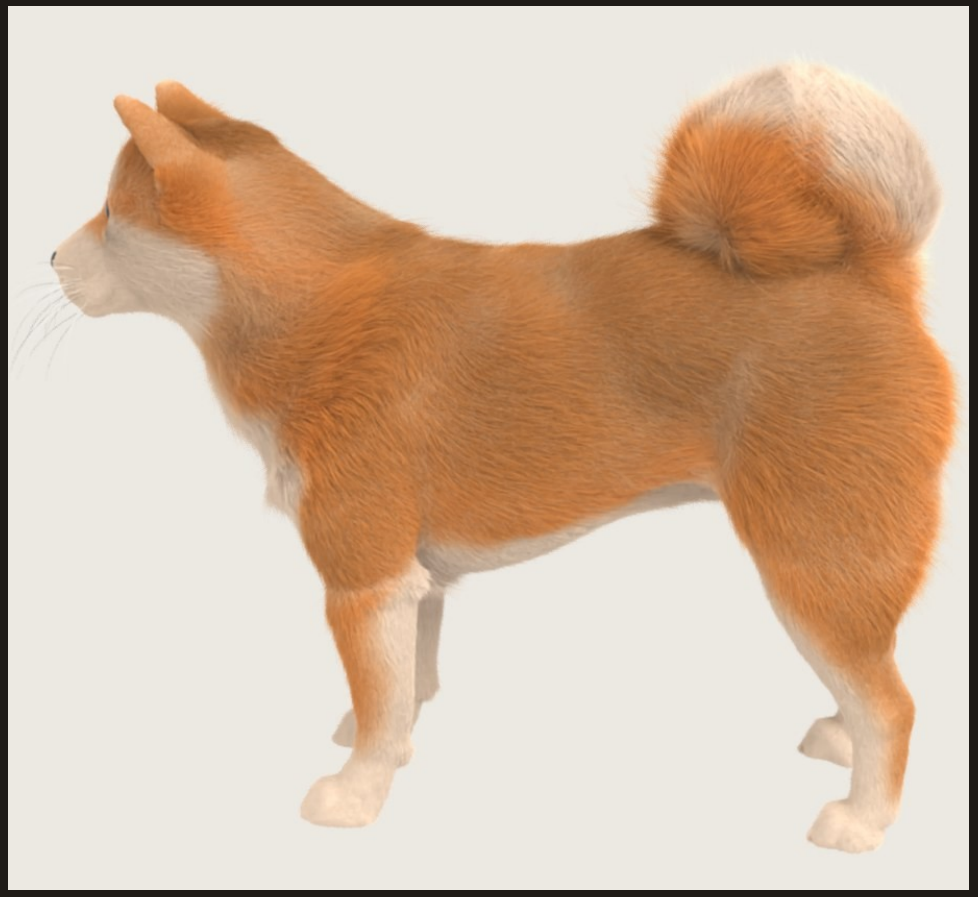
PROJECT REPORT • NOVEMBER 2026

From three photos to four million hair strands

A general, runnable animal-modeling pipeline built on diffusion models



Code: github.com/CCdydy/diffusion_pet • Weights: huggingface.co/zengzzy/diffusion_pet • Generated video: MiniMax H3

THE GOAL

A usable, general, diffusion-based pipeline for modeling animals

Input

Three ordinary photos of one animal. No camera calibration, no multi-camera rig.

Output

An explicit 3D groom: millions of strand curves on a closed body mesh, ready to render and edit.

Requirements

Runs on a new breed by swapping one config file. About an hour on one workstation. No per-animal training from scratch.

THE PIPELINE

Five stages, three diffusion models

1

White body mesh

Fur-less video from the photos, then a closed mesh

H3 + Hunyuan3D

2

Furry videos

Mesh depth drives the video; every frame has a known camera

MiniMax H3

3

Case

Cameras, fur masks, direction maps, alignment check

Geometry tools

4

Fur prior

Strand shape and length per surface patch, one sampling pass

v18 (ours)

5

Coat and color

Two-layer growth, breed table, a few fixed coat colors

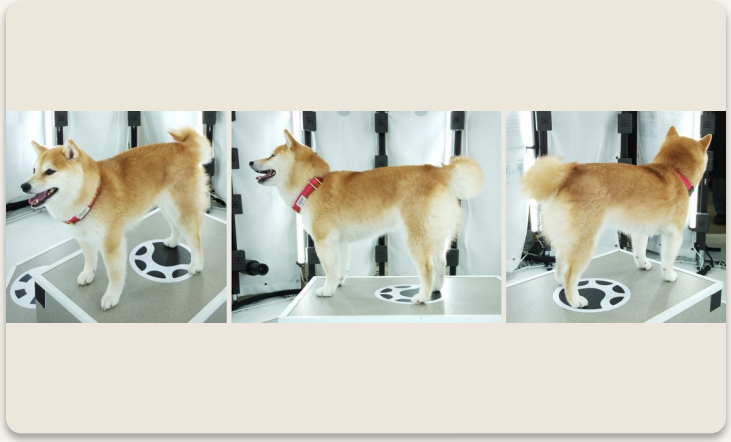
Growth tools

Orange outline = a diffusion model does the work. Only one config file changes between animals: species, ears, tail, coat description, coat table and colors.

H3 = MiniMax H3 video diffusion • Hunyuan3D = Hunyuan3D-2mv multi-view 3D diffusion • v18 = our geometry-conditioned fur prior

STAGE 1 • WHITE BODY MESH

Shave the dog in video, then lift it to 3D



Three photos. Front-left, left, back-right; no calibration.



H3 fur-less video. Prompted: no fur, matte white, mouth closed, paws still.

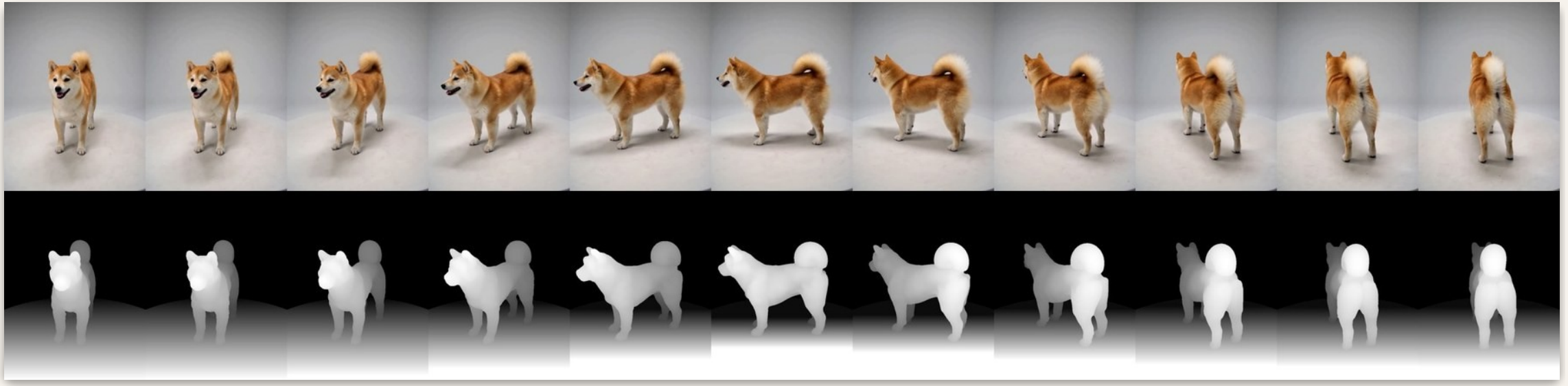


Hunyuan3D-2mv. First and last frames in, one closed, cleaned mesh out (about 130k vertices).

About 3 minutes • the mesh is then scaled to metres and turned so the head points forward, from one height value in the species config

STAGE 2 • FURRY VIDEOS WITH KNOWN CAMERAS

The mesh drives the video, so every frame has a camera



Control. Mesh depth guides H3 for the first 40% of denoising; then the fur grows past the silhouette.

Coverage. 3 orbits and 19 close-ups: muzzle, cheeks, ears, chest, legs, paws, tail, back.

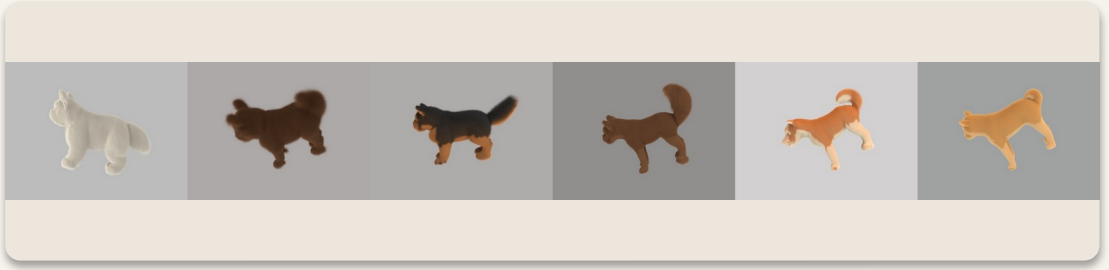
Output. 1,364 frames aligned with the mesh, plus fur masks and direction maps.

22 clips of 124 frames • 30–36 minutes on two GPUs (RTX 4090 + RTX 6000 Ada) • coat words and per-part fur come from the species config

STAGE 4 • FUR DIFFUSION PRIOR (OURS)

A diffusion model that knows how fur grows on curved skin

- For each surface patch, samples the strands that grow there along the given flow: how they rise, droop and clump.
- Conditioned on multi-view image features and the local skin shape: height, normals, gravity.
- Trained only on synthetic dogs: random SMAL bodies with realistic coat types; 135 dogs, 8,640 patches.
- One sampling pass of 10 steps replaces per-animal optimization.



Held-out test bodies (1,280 patches)	Shape error ↓
v17, flat-patch prior	0.430
Procedural growth, same parameters	0.220
v18 without geometry	0.181
v18	**0.163**

Shape error = median per-patch point error ÷ strand length, against the synthetic ground truth

STAGES 3 AND 5 • FROM VIDEO TO A GROOMED COAT

Video gives the flow, diffusion the length, the config the style

Flow

Solved in closed form from all 1,364 frames' direction maps, then smoothed on the mesh. No swirls, no seams.

Length

From the diffusion prior, each strand scaled so the coat matches the fur silhouettes in the video.

Two-layer coat

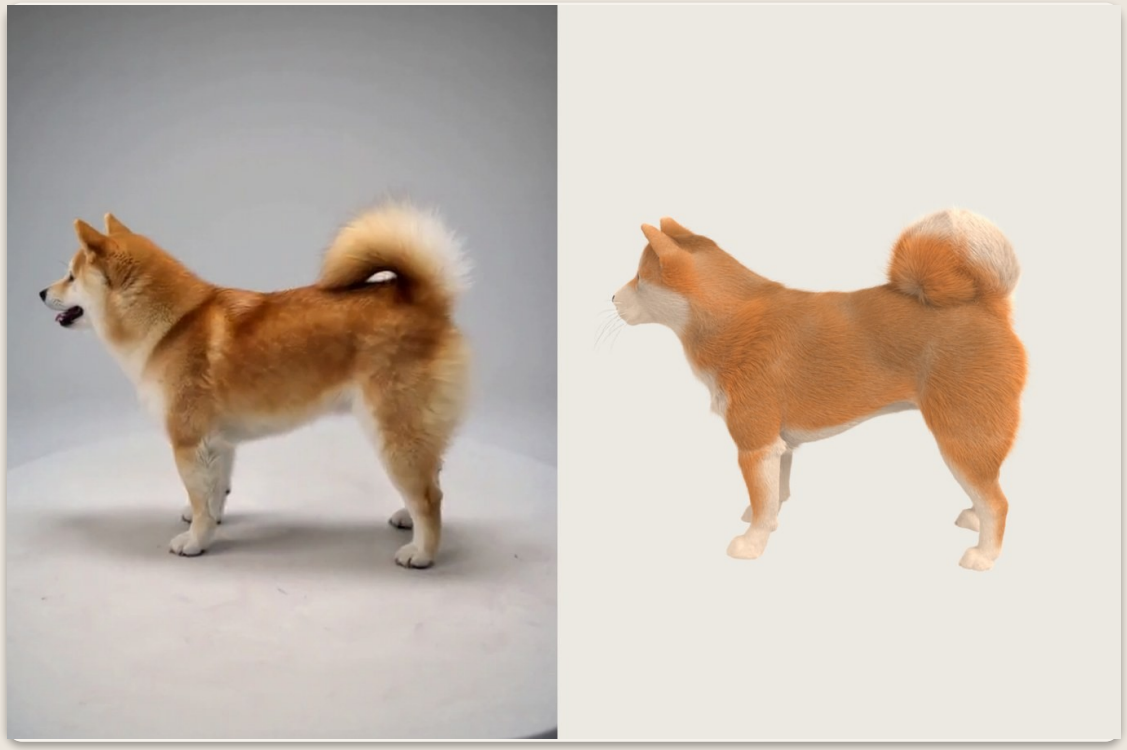
Guard hairs that rise at the root and lie down along the flow, over a short, lofty undercoat; per-part table in the config.

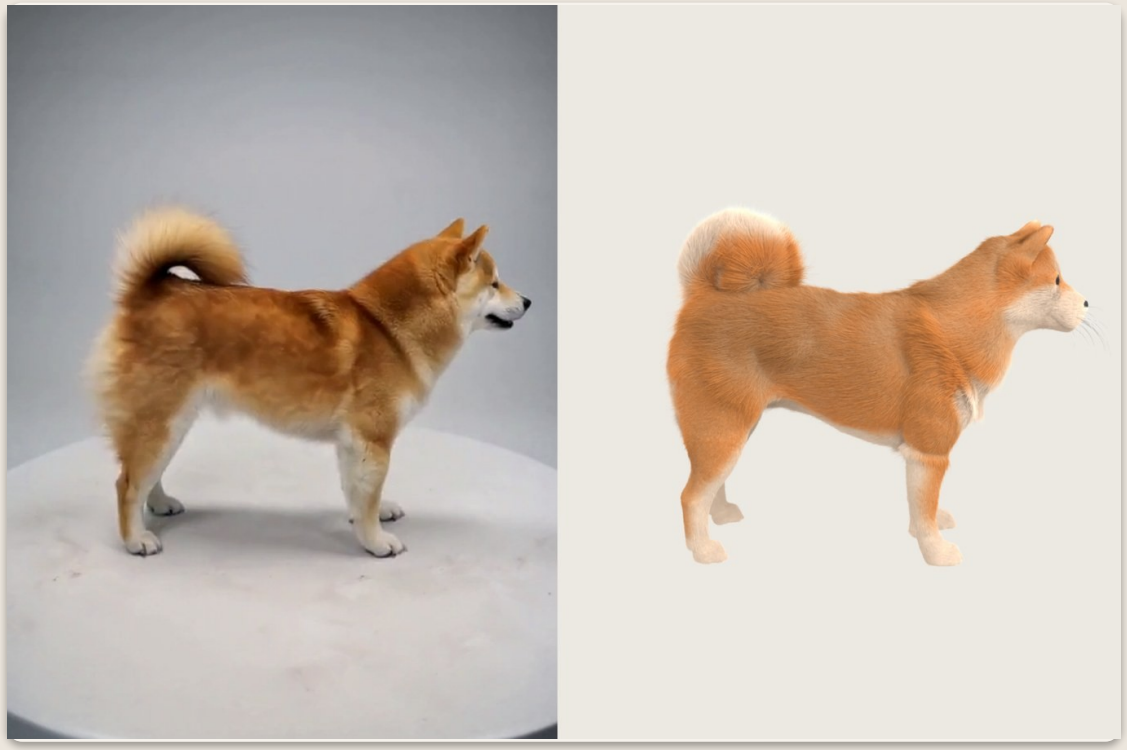
Color

A few fixed coat colors mixed by smooth weights read from the video, so no per-strand photo noise. Eyes and whiskers added last.

RESULT • SHIBA INU

H3 frame and our strands, from the same camera





In each pair: left = H3 frame, right = render of the reconstructed groom (Blender Cycles, hair shader) • red guard hairs over a cream undercoat

RESULT • CLOSE-UPS

Close-ups on frames the reconstruction never used



Back



Chest



Thigh



Tail

In every clip, frames 0, 4, 8 … feed the reconstruction and frames 2, 6, 10 … are held out; these pairs are held-out frames. Each image: left = H3 frame, right = our render.

Strand radius 150 µm (guard) / 90 µm (undercoat) at render scale • whiskers mixed dark and white

THE FINAL MODEL

4,000,050 strands

4 million fur strands (53% guard hair, 47% undercoat) plus 50 whiskers, each an explicit 3D curve

80 M

curve points, 20 per strand

54 km

of hair in total

11.6 mm

median strand length

129 k

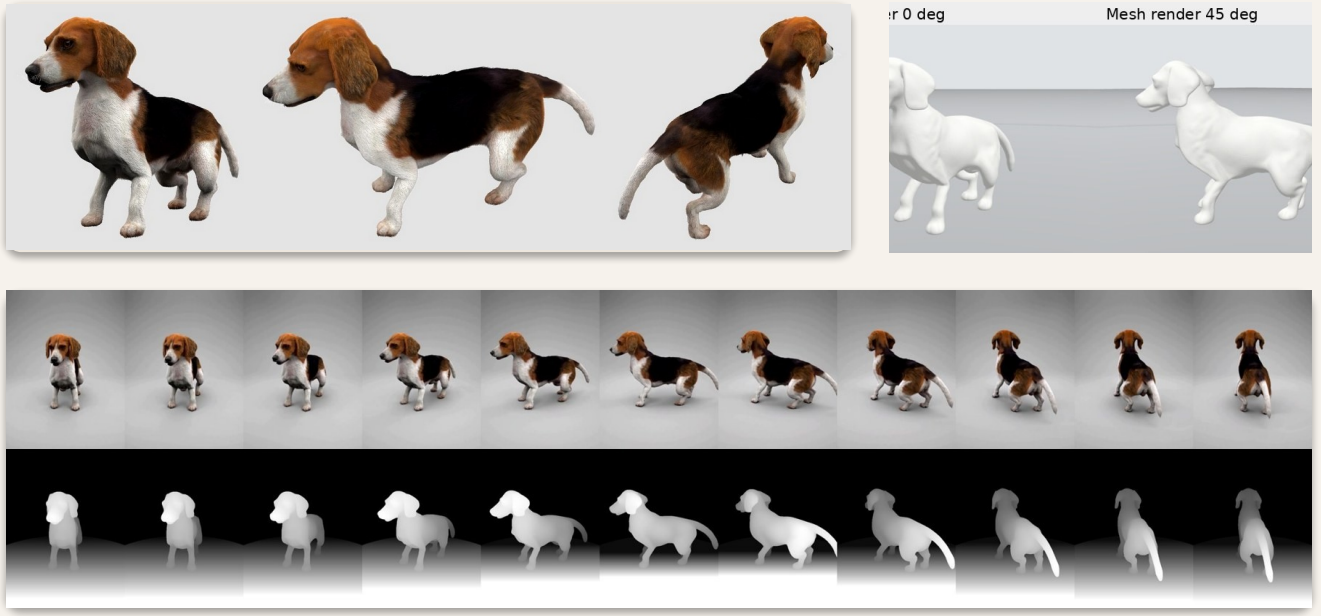
vertices on the closed skin mesh

GENERALITY • BEAGLE

Same program, one new config file

- Floppy ears and a straight tail: different rules place the close-ups and label the parts.
- Coat words for H3: black saddle, tan, white.
- Coat table: short, smooth, lies flat, almost no clumping.
- Three coat colors instead of two.

Nothing else changed. The run exposed four bugs that only a second animal could find; all are fixed in the shared code.

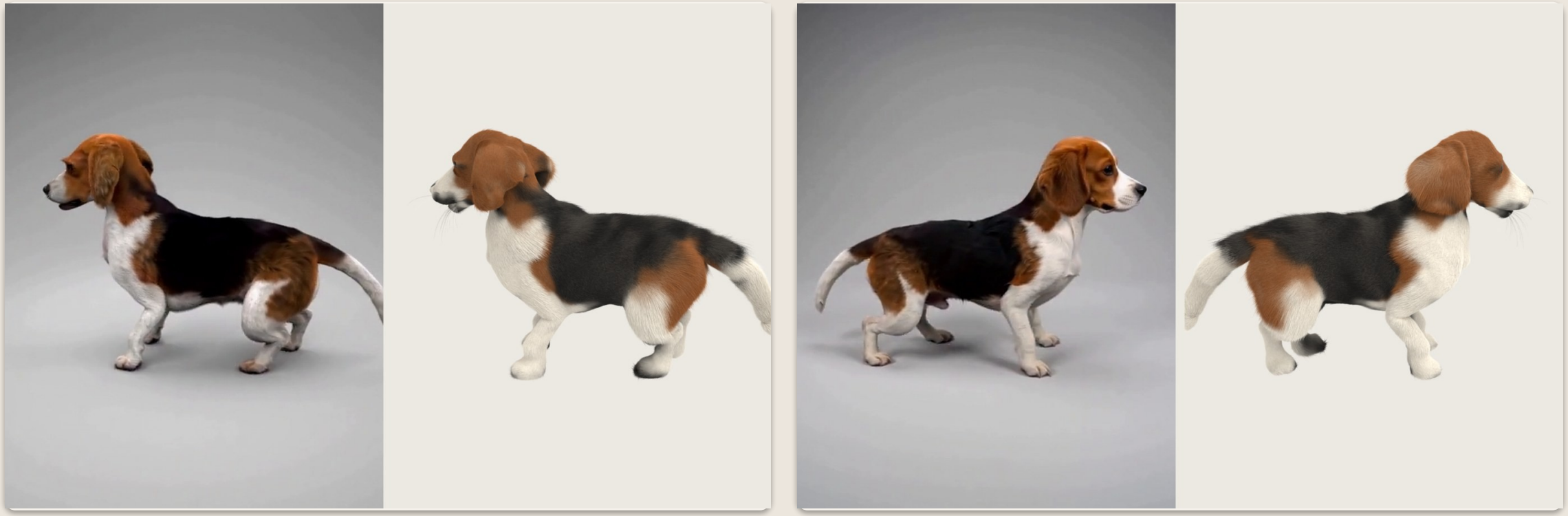


Photos → white mesh → H3 orbit with the mesh depth as control (bottom row)

Beagle photos: public ARTEMIS renders • fixes: SAM2 QC on close-ups, NaN guard in the flow solver, eye detection next to dark coat, root-to-patch lookup

RESULT • BEAGLE

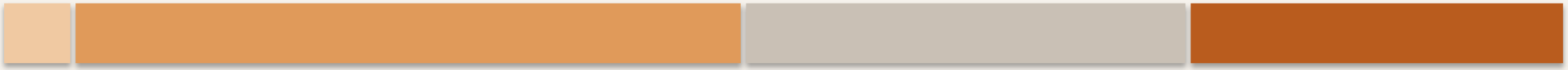
Black saddle, tan and white, from the same camera



In each pair: left = H3 frame, right = our render • 4,000,041 strands (68% guard hair, median 11.8 mm) • black 36%, tan 26%, white 38% of the skin

TIME

About 70 minutes of compute, from photos to groom



Furry videos • 30 min
22 H3 clips, two GPUs

Case • 20 min
SAM2 masks on 1,364 frames

Fur • 17 min
v18 pass 5.5 • coat 0.3

White body mesh: 3.2 min (H3 fur-less video 2.3 + Hunyuan3D 0.8)

Fur reconstruction alone, same scale

Ours: about 17 min

Per-animal optimization from scratch (NeuralFur): about 10 hours

Beagle, packaged program, RTX 4090 + RTX 6000 Ada • Shiba Inu (earlier route with the v17 prior): 78 min • rendering not included • NeuralFur as reported (A100)

USING IT

A new animal is one config file

```
bash pipeline/run_animal.sh <photos> pipeline/species/beagle.json <out>
```

In the species config	Shiba Inu	Beagle
Ears • tail	erect • curled	floppy • straight
Coat description for H3	red guard hair, cream undercoat	black saddle, tan, white
Coat table (per body part)	fluffy double coat, clumped	short, smooth, lies flat
Coat colors	2 (red, cream)	3 (black, tan, white)
Standing height	0.47 m	0.48 m

Code: github.com/CCdydy/diffusion_pet • weights: huggingface.co/zengzzy/diffusion_pet • two GPUs (ours: RTX 4090 + RTX 6000 Ada), ComfyUI with H3 and Hunyuan3D-2mv

LIMITS

What is not solved yet

- The face is the weakest region: eyes are round, without lids, and the eye opening is rough.
- Shape errors in the white mesh carry through, for example the volume of a curled tail.
- The coat table is still set by hand per breed.
- Commercial use of H3 output needs a MiniMax license.

NEXT

Where it goes from here

- Let the diffusion prior also predict the coat table, so a new breed needs no hand-set values.
- Close-up face clips at higher resolution for eyes, nose and whiskers.
- More animals: long-haired dogs and cats.
- Strand-level editing: change length or color of one body part and regrow.